3-Bit Digital-to-Analog Converter

ELECENG 2EI4

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Summary

This project involved the design, simulation, and implementation of a 3-bit Digital-to-Analog Converter (DAC) with a 5 V full-scale output using a weighted resistor topology and an inverting summing amplifier. The circuit incorporated voltage follower buffers to minimize current draw from the digital inputs and was validated through LTspice simulations and experimental testing. Measured performance showed low offset and gain errors along with differential non-linearity within ±1 LSB, confirming accurate operation and monotonic behavior without missing codes.

Design

Topology Selection

A weighted resistor DAC was selected as the design topology. In this approach, each bit controls a current through a resistor weighted in proportion to the binary significance of that bit. The MSB drives the smallest resistor (highest current) and the LSB drives the largest resistor (lowest current). The currents sum at the virtual ground of an inverting op-amp, which converts the total current into an output voltage. This topology was chosen for its simplicity and ease of analysis for a 3-bit implementation, where the required resistor values (10 kΩ, 20 kΩ, 40 kΩ) fall within a manageable 4:1 range and are available as standard values in the component kit.

Circuit Diagram

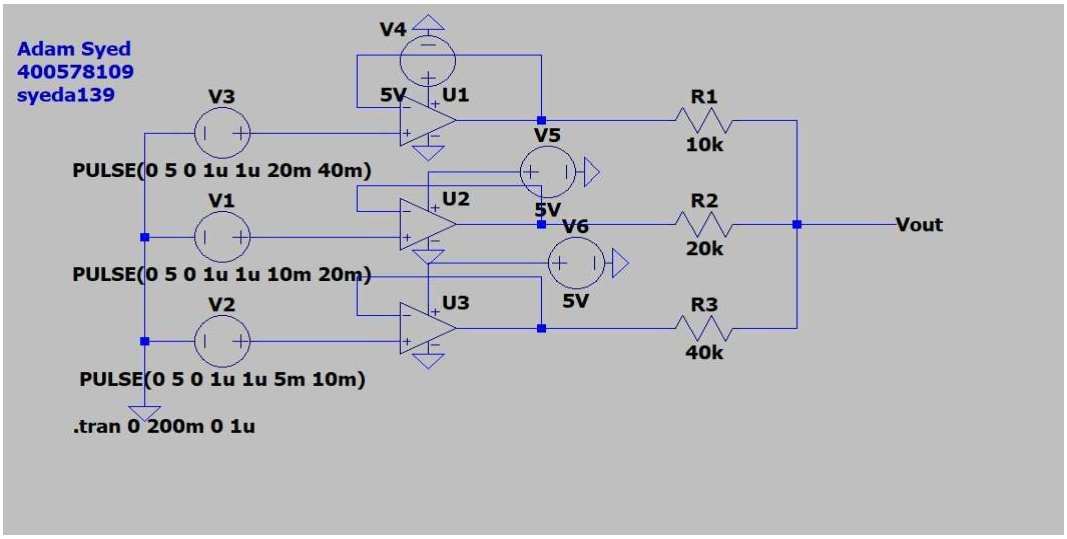


Figure 1. LTspice schematic

Description of Circuit Operation

The circuit operates in three stages. First, each bit voltage (B2, B1, B0) is applied to the non-inverting input of a unity-gain buffer. The buffers are meant to have a high impedance to have a near zero current from the digital lines while reproducing the bit voltage at their output. This

fully satisfies the constraint that current drawn from the bit lines must not exceed 1 μA as the maximum current is 5 nA.

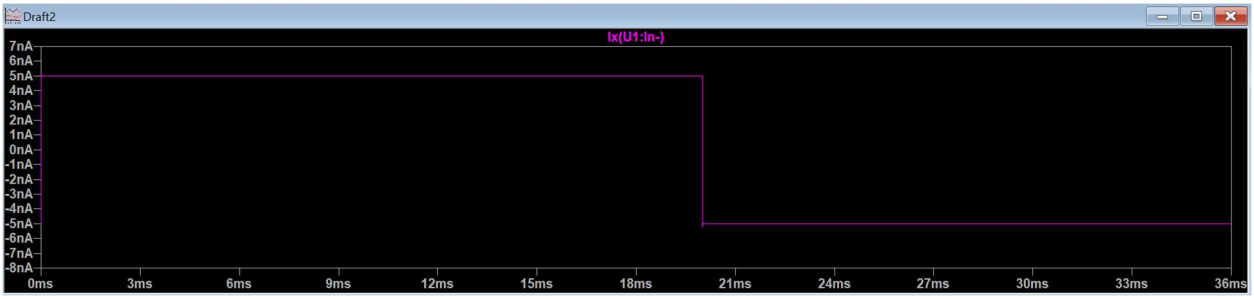


Figure 2. Buffer Input Current Simulation

Second, the buffered bit voltages drive the weighted resistor network. B2 connects through R1 = 10 kΩ, B1 through R2 = 20 kΩ, and B0 through R3 = 40 kΩ. The 1:2:4 ratio means B2 contributes twice the current of B1 and four times the current of B0, correctly encoding the binary weights.

Third, the weighted currents are summed at the inverting input of the output op-amp. With feedback resistor Rf, the output is given by the equation below. This

$$V_{out} = -R_t \times (Vb_2/R1 + Vb_1/R2 + Vb_0/R3)$$

Since the summing amplifier inverts the signal, we use the allowed inversion convention where 0 V represents logic-1 and 5 V represents logic-0. This way, as the input code goes from 000 to 111, the output voltage increases smoothly and stays monotonic.

The weighted resistor DAC topology and inverting summing amplifier design are based on information found within section 1.3 of the textbook along with lecture slides from COMPENG 2CF3.

Design Calculations

The ideal LSB step size is determined by the full-scale voltage and bit count:

$$Vlsb = Vfs / 2^n = 5 / 8 = 0.625 \text{ V}$$

The maximum output at code 111 should equal 7 × 0.625 = 4.375 V. Solving for the required feedback resistor Rf with all bit inputs at V_{hi} = 5 V:

$$4.375 = Rf \times 5 \times (1/10k + 1/20k + 1/40k) = R_t \times 5 \times (7/40k)$$

$$Rf = 5{,}000\ \Omega = 5\ k\Omega$$

The inverting amplifier was not implemented in the LTspice schematic due to it being able to have an input full scale voltage of 5V. However, on the AD3, the maximum full scale voltage is 3.3V, therefore it needed to be amplified. The following circuit was used:

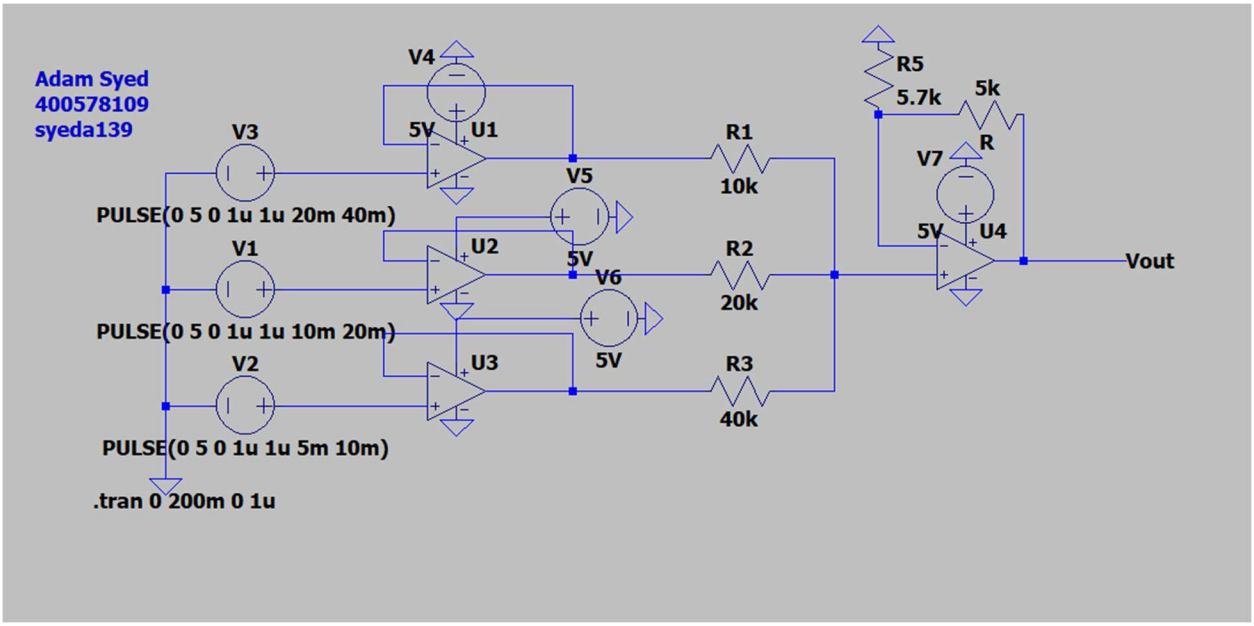


Figure 3. Implemented circuit schematic

Measurement and Analysis

i. Photograph of Circuit

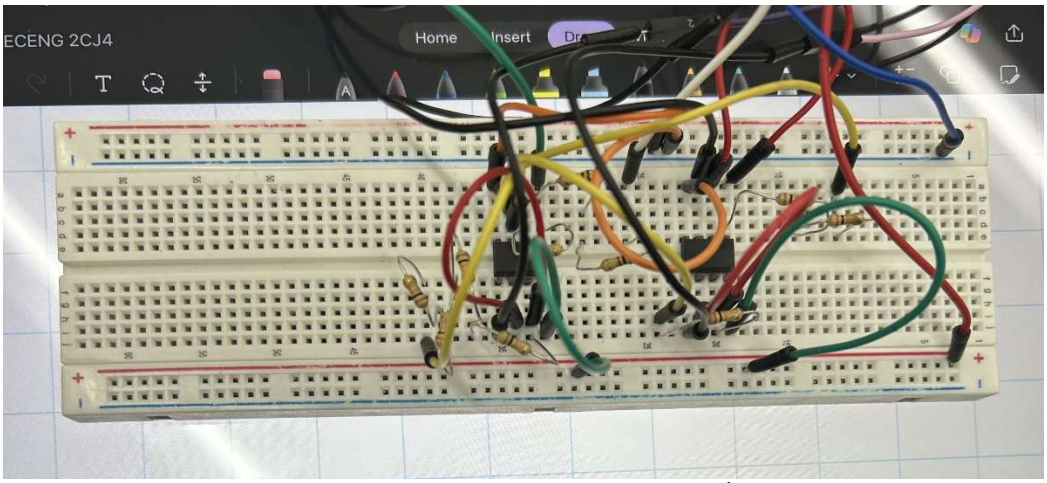


Figure 4. Physical Circuit taken on April 11th, 2026

ii. Generation of Digital Inputs

Three digital bit inputs (B2, B1, B0) were required to implement all eight 3-bit input codes. In the LTspice simulation, three independent pulse voltage sources (V1, V2, V3) were used, each switching between 0 V and 5 V with different periods so that all eight binary codes were produced in sequence over one full cycle:

- **V3 (B2, MSB):** PULSE(0 5 0 1u 1u 20m 40m) — period 40 ms, on for 20 ms
- **V1 (B1):** PULSE(0 5 0 1u 1u 10m 20m) — period 20 ms, on for 10 ms
- **V2 (B0, LSB):** PULSE(0 5 0 1u 1u 5m 10m) — period 10 ms, on for 5 ms

These bits all toggle in the LTspice simulation to decrement from 111 to 000. In the physical circuit, three digital output channels from the AD3 were used to set each 3-bit code manually, and the output voltage was measured using the scope for each code.

iii. Transfer Characteristics

The full transfer characteristic was obtained by recording the output voltage Vout for each of the eight digital input codes. Figure 4 shows the simulation waveform from LTspice, and Figure 5 and 6 shows the corresponding waveform captured on the physical circuit using the oscilloscope on the AD3. It is reversed, compared to the LTSpice simulation waveform.

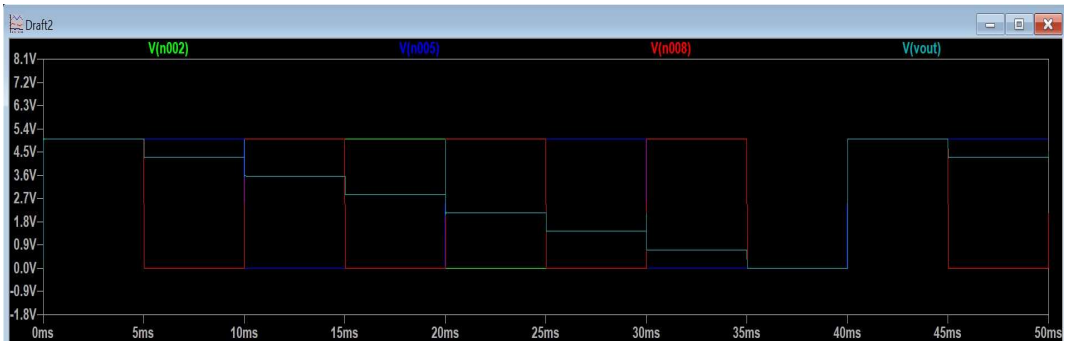


Figure 5. LTspice simulation

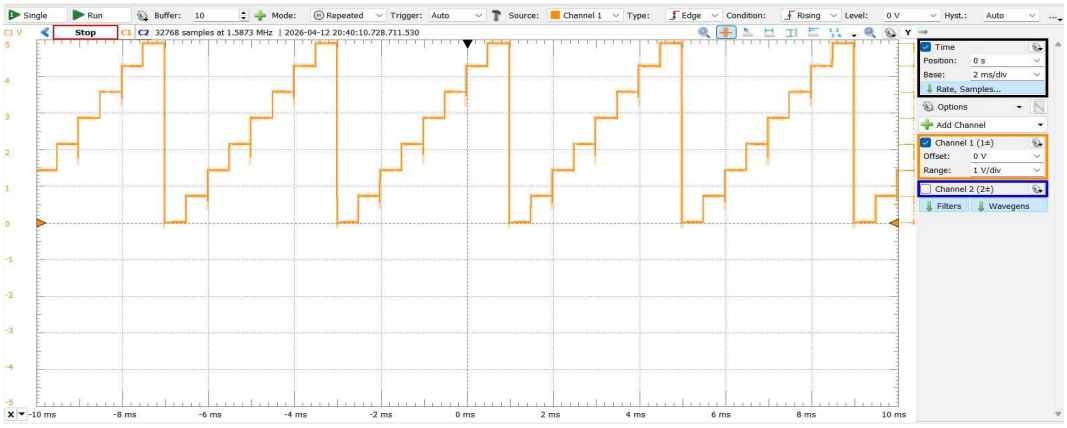


Figure 6. Waveforms scope without cursors

The staircase waveforms above clearly shows eight distinct and increasing output voltage levels, confirming correct DAC operation. The output starts near 0 V for code 000 and reaches approximately 4.93 V for code 111. The output levels were precisely measured using eight Y-

axis cursors in WaveForms, shown in Figure 6. They are compared to ideal values in Table 1 below.

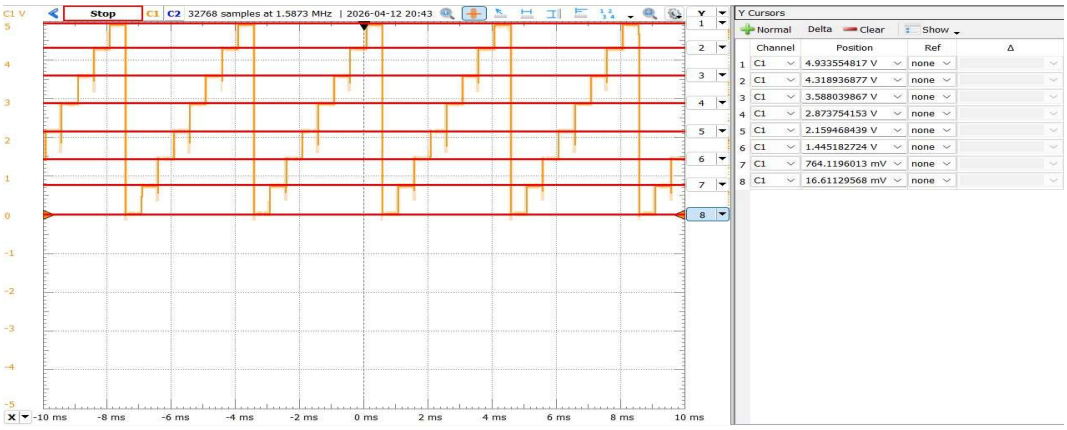


Figure 7. Waveforms scope with cursors

B2	B1	B0	Decimal	Ideal Vout (V)	Measured Vout (V)
0	0	0	0	0.000	0.0166
0	0	1	1	0.625	0.7641
0	1	0	2	1.250	1.4452
0	1	1	3	1.875	2.1595
1	0	0	4	2.500	2.8738
1	0	1	5	3.125	3.5880
1	1	0	6	3.750	4.3189
1	1	1	7	4.375	4.9336

Table 1. Ideal versus measured transfer characteristics.

iv. Gain Error

The gain error is the difference between the actual and ideal full-scale output voltage, expressed in LSB. It represents a scaling error in the slope of the transfer characteristic: a positive gain error means the output is larger than expected, while a negative error means it is smaller.
To measure the gain error, the output at the highest input code (111) is compared to the ideal value:

$$\text{Gain Error} = \frac{V_{\text{out, actual}} - V_{\text{out, ideal}}}{V_{\text{LSB}}}$$

Using the measured value 4.9336 Vand ideal value 4.375 V:

$$\text{Gain Error} = \frac{4.933 - 4.375}{0.625} = +0.893 \text{ LSB}$$

The positive gain error indicates that the output is consistently higher than ideal across all input codes. This suggests the effective gain of the summing amplifier is slightly larger than designed. A likely cause is the use of a 4.99 kΩ feedback resistor instead of the ideal 5 kΩ, along with resistor tolerances in the weighted network, resulting in an overall increase in gain.

v. Maximum Differential Non-Linearity (DNL)

The differential non-linearity (DNL) for a given code transition is the deviation of the actual output step size from the ideal step size of one LSB (0.625 V), expressed in LSBs:

$$DNL_n = (V_{out}(n) - V_{out}(n-1) - Vlsb) / Vlsb$$

The step sizes were calculated from the cursor readings in Figure 4. Each step is the voltage difference between adjacent cursor positions. Table 2 below shows the step size and DNL for every code transition.

Transition	Step Size (V)	Ideal Step (V)	DNL (LSB)
0 → 1 (000 → 001)	0.7475	0.625	+0.1960
1 → 2 (001 → 010)	0.6811	0.625	+0.0897
2 → 3 (010 → 011)	0.7143	0.625	+0.1429
3 → 4 (011 → 100)	0.7143	0.625	+0.1429
4 → 5 (100 → 101)	0.7142	0.625	+0.1427
5 → 6 (101 → 110)	0.7309	0.625	+0.1694
6 → 7 (110 → 111)	0.6147	0.625	-0.0165

Table 2. Step sizes and DNL

The maximum DNL is +0.196 LSB, occurring at the 000 to 001 transition (step size 0.7475 V vs. ideal 0.625 V). Since all DNL values are well within ±1 LSB, the DAC is confirmed to be monotonic with no missing output codes. The relatively uniform positive DNL across most transitions (approximately +0.14 to +0.17 LSB) is consistent with the positive gain error: the effective LSB step is proportionally larger than 0.625 V at all transitions due to the higher-than-designed amplifier gain.

vi. Offset Error

The offset error is the output voltage when all digital inputs are set to zero (code 000). Ideally this should be exactly 0 V. It represents a fixed additive shift of the entire transfer curve and originates from the input offset voltage of the output op-amp U4.

Measurement: With all bit inputs at 0 V (code 000), the output voltage was read directly from cursor C8 in Figure 4.

$$\text{Offset Error} = 16.61 \text{ mV} = 16.61 \text{ mV} / 625 \text{ mV} = 0.027 \text{ LSB}$$

The offset error is 16.61 mV, which is 0.027 LSB. This value is extremely small and well below the 0.5 LSB threshold that would indicate a meaningful error. This is consistent with op-amp's low input voltage offset. The 16.61 mV observed in the physical circuit is likely attributable to a combination of the op-amp offset voltage, small resistor mismatch creating a non-zero current at zero input, and measurement noise picked up by the scope.

Discussion

Gain error (+0.894 LSB):

The positive gain error indicates the output is consistently higher than ideal, meaning the overall gain of the summing amplifier is slightly larger than designed. This is mainly due to resistor tolerances in the feedback and weighted resistors, which scale all output values proportionally and result in a slope error.

Maximum DNL (+0.196 LSB):

The DNL error is caused by small mismatches in the resistor ratios, which affect the size of individual output steps. The largest deviation occurs at the LSB transition, suggesting that branch contributes slightly more than expected. Since all DNL values are within ±1 LSB, the DAC remains monotonic.

Offset error (16.61 mV, 0.027 LSB):

The offset error comes from the op-amp's internal offset voltage and minor measurement imperfections. Even when all inputs are zero, these effects cause a small non-zero output.

Design Choice and Alternate Architecture:

The weighted resistor DAC was chosen because it is simple and easy to implement for a 3-bit system. However, it does not scale well since higher resolutions require a large range of resistor values. The R-2R ladder is a better alternative, as it uses only two resistor values and is easier to scale and match accurately, making it more suitable for practical designs.